Domain-Aware Submodular Optimization for Data Selection

Twumasi Mensah-Boateng
CSCE 6270 - Advanced Human/Machine Interaction
Computer Science and Engineering
University of North Texas
twumasimensah-boateng@my.unt.edu

Abstract

Domain adaptation remains a significant challenge in natural language processing, particularly when labeled data in target domains is scarce. In this paper, we propose DASS (Domain-Aware Submodular Selection), a framework that leverages submodular optimization to select representative samples from both target and source domains for efficient model adaptation. Our two-stage approach first identifies diverse representative samples from the target domain using facility location functions, then selects complementary examples from the source domain using submodular mutual information, creating a balanced dataset that captures target domain characteristics while leveraging source domain knowledge.

Through experiments on sentiment analysis tasks using Amazon reviews (source domain) and Yelp reviews (target domain), we demonstrate that DASS outperforms random selection baselines. Our analysis reveals that data points selected using our submodular approach align more closely with the target domain distribution, confirming that quality and relevance of training data can be more important than quantity. These results establish submodular information measures as a promising foundation for domain adaptation in natural language processing tasks, particularly in low-resource scenarios.

1 Introduction

Transformer-based language models (LLMs) (Vaswani et al., 2017) such as BERT (Devlin et al., 2019), LLaMA (Touvron et al., 2023) and GPT (Radford et al., 2018) have transformed natural language processing and artificial intelligence. Trained on vast datasets encompassing much of the internet, these models demonstrate remarkable capabilities in few-shot and zero-shot settings (Brown et al., 2020). However, to maximize their effectiveness for specific applications, fine-tuning on small, well-curated domain-specific datasets yields superior results (Wei et al., 2022; Chung et al., 2022).

Despite the general abundance of data, specialized domains often suffer from data scarcity, hindering the development of effective models. Traditional solutions to this problem—such as generating synthetic data or manually labeling datasets (Marcus et al., 1993; Bowman et al., 2015)—introduce challenges including domain shift, annotation costs, and privacy concerns.

To address these limitations, researchers have turned to domain adaptation, a specialized form of transfer learning (Wilson and Cook, 2020; Ben-David et al., 2010). This approach leverages models trained on source domains with abundant labeled data to perform effectively in target domains with limited labeled data (Guo and Yu, 2022; Ke et al., 2022; Ala et al., 2021; Saunders, 2022; Costa et al., 2022).

In this paper, we address domain adaptation with techniques from active learning (Settles, 2009) and submodular optimization (Fujishige, 2005; Krause and Golovin, 2014). Specifically, we investigate how selecting a representative subset of the target domain can guide more effective sample selection in the source domain. We formulate this challenge using submodular information measures (Iyer et al., 2021), which provide a principled framework for quantifying the informativeness of data subsets. Through extensive experimentation, we demonstrate that our approach based on submodular information measures consistently outperforms baseline methods.

Our contributions include: (1) a two-stage framework for domain adaptation that first identifies representative target domain samples and then selects complementary source domain instances based on submodular mutual information, and (2) empirical evidence across using sentiment analysis task demonstrating that this framework significantly improves performance when adapting models to new

domains with limited data.

This work addresses the following research questions:

- **R.Q.1.** Which data points from the target domain should be annotated?
- **R.Q.2.** How do features in the annotated samples guide selection from the source domain?
- **R.Q.3.** How can we select a representative subset from the source domain that closely relates to the annotated target domain data?

2 Methodology

We propose a submodular data selection framework for efficient domain adaptation based on submodular mutual information, enabling optimal knowledge transfer from a large-scale source domain to low-resource target domains. Our approach selectively identifies the most informative examples from the source domain, maximizing performance on target domains while minimizing computational costs.

2.1 Overview of the DASS Framework

The Domain-Aware Submodular Selection (DASS) framework provides a principled approach to domain adaptation by strategically selecting a small number of annotated target domain samples and complementary source domain examples. As illustrated in Figure 1, our approach operates in two distinct stages.

In the first stage, we address the challenge of identifying which data points from the target domain should be annotated (R.Q.1). Instead of random selection, we employ a facility location function to identify a diverse and representative subset of the target domain. This subset provides maximum coverage of the target distribution with minimal annotation cost, ensuring that the selected samples capture the key characteristics of the target domain.

The second stage focuses on leveraging these annotated target samples to guide the selection of the most relevant source domain examples (R.Q.3). Using submodular mutual information, we identify source domain instances that share similar features with the annotated target samples while maintaining diversity. This ensures that the selected source domain examples contain knowledge that can effectively transfer to the target domain, addressing the challenge of domain shift. By balancing relevance to the target domain with diversity in the source selection, our framework creates a training dataset that maximizes model performance while minimizing computational costs.

The DASS framework offers several advantages over previous approaches: (1) it reduces annotation costs by identifying the most informative target domain samples, (2) it decreases training data size by selecting only the most relevant source domain examples, and (3) it provides theoretical guarantees through the properties of submodular optimization. The framework is model-agnostic and can be applied to various NLP tasks, making it a versatile solution for domain adaptation challenges in low-resource scenarios.

2.2 Problem Formulation

Let $\mathcal{D}_S = \{(x_i^s, y_i^s)\}_{i=1}^{N_S}$ represent our labeled source domain dataset with N_S examples, and $\mathcal{D}_T = \{(x_j^t)\}_{j=1}^{N_T}$ represent our unlabeled target domain dataset with N_T examples. Additionally, we may annotate a small portion of the target domain dataset $\mathcal{T} = \{(x_j^t, y_j^t)\}_{j=1}^{N_t}$, where $N_t \ll N_T$.

Our goal is to select a subset $S \subset \mathcal{D}_S$ such that $|S| \ll |\mathcal{D}_S|$ and a model trained on $\mathcal{D}_{\text{final}} = S \oplus \mathcal{T}$ performs comparably to a model trained on a fully labeled target dataset $\mathcal{D}_T^{\text{labeled}} = \{(x_k^t, y_k^t)\}_{k=1}^{N_T}$ if we had access to all the labels.

2.3 Submodular Selection Framework

Submodular functions provide a framework for balancing relevance, diversity, and informativeness in data selection. A function $f : 2^V \to \mathbb{R}$ is submodular if for all subsets $A \subseteq B \subset V$ and all elements $v \in V \setminus B$, we have:

$$f(A \cup \{v\}) - f(A) \geq f(B \cup \{v\}) - f(B) \quad (1)$$

This property captures the diminishing returns nature of adding examples to a dataset.

Target Domain Unlabeled Data Selection

For selecting representative and diverse unlabeled examples from the target domain for annotation, we use the facility location function:

$$f_{fl}(\mathcal{T}) = \sum_{i \in \mathcal{D}_\mathcal{T}} \max_{j \in \mathcal{T}} S_{i,j} \quad (2)$$

Using the above formulation, we now have the objective function. Thus we want to find the representative subset:

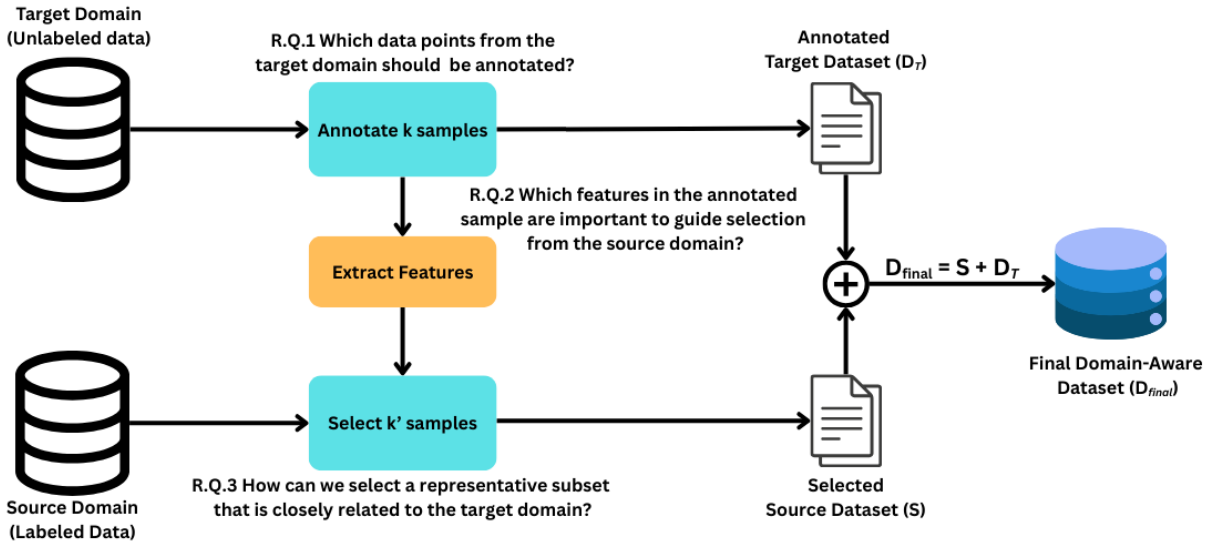


Figure 1: The DASS Framework

$$\mathcal{T} = \underset{\mathcal{T} \subseteq \mathcal{D}_T : |\mathcal{T}| \leq k_1}{\operatorname{argmax}} f_{fl}(\mathcal{T}) \tag{3}$$

where k_1 is the number of data points from the target domain that will be annotated.

Source Domain Data Selection

To select a subset of data points from the source domain that are more similar to the annotated dataset from the target domain, we turn to information theory. Specifically, we use the mutual information between two random variables which is defined as in terms of entropy,

$$I(X;Y) = H(X) - H(X|Y) \tag{4}$$

Similarly, the submodular mutual information between two sets $A, Q \subseteq V$ denoted by $I_f(A;Q)$ is defined as $I_f(A;Q) \triangleq f(A) + f(Q) - f(A \cup Q)$ (Kothawade et al., 2022; Iyer et al., 2021). This measures the similarity between Q and A, where Q is the query set. As such, using the more scalable Facility Location Query Mutual Information function from (Kothawade et al., 2022), we define our mutual information function as

$$f(S, \mathcal{T}) = \sum_{i \in \mathcal{T}} \max_{j \in S} S_{i,j} + \eta \sum_{i \in S} \max_{j \in \mathcal{T}} S_{i,j} \tag{5}$$

where $\eta \in [0, 1]$ controls the balance between query relevance and diversity, typically set to $\eta = 1.0$ in our experiments.

Therefore the objective is to find a subset of the source domain (S) which maximizes the mutual information with the annotated data points from the target domain ($\mathcal{T}$):

$$S = \underset{S \subseteq \mathcal{D}_S : |S| \leq k_2}{\operatorname{argmax}} f(S; \mathcal{T}) \tag{6}$$

where $k_2 = \beta N_S$ with $\beta \in [0, 1]$ based on task requirements and computational constraints.

Intuition Behind Submodular Functions

Our approach leverages two specific submodular functions that provide complementary benefits for domain adaptation. The facility location function used for target domain selection exhibits an inherent coverage property that ensures diversity while maximizing representativeness. Intuitively, it selects points that act as "facilities" serving all data points in the target domain—each selected point represents a neighborhood of similar examples. This property is particularly valuable for obtaining a comprehensive view of the target domain with minimal annotation cost.

For source domain selection, we employ submodular mutual information because it naturally quantifies the information shared between the annotated target subset and candidate source points. This measure balances two critical factors: (1) relevance to the target domain examples, ensuring knowledge transfer is focused on features present in the target domain, and (2) diversity within the selected source subset, preventing redundancy. The parameter η allows us to adjust this trade-off, with higher values favoring diversity when the domains are relatively similar, and lower values prioritiz-

ing relevance when domains are highly divergent. This principled information-theoretic approach ensures that each selected source example provides maximum benefit for the adaptation process.

Optimization Algorithm

Since maximizing submodular functions is NP-hard, we employ the greedy algorithm (Algorithm 1) with lazy evaluations (Mirzasoleiman et al., 2015) which provides a $(1 - 1/e)$ approximation guarantee for monotone submodular functions. The algorithm iteratively selects the element with maximum marginal gain:

$$v^* = \underset{v \in V \setminus S}{\operatorname{argmax}} f(S \cup \{v\}) - f(S) \qquad (7)$$

The Final Dataset

The final dataset for the task is $\mathcal{D}_{final} = S \oplus \mathcal{T}$, where $\mathcal{T}$ is a small manually labeled dataset from the target domain used to guide the selection of S from the source domain. This combined dataset achieves an optimal balance between target domain relevance and source domain diversity, enabling efficient domain adaptation with minimal annotation cost.

Algorithm 1 DASS: Domain-Aware Submodular Selection Algorithm

Require: Target corpus $\mathcal{D}_{\mathcal{T}}$, Source corpus $\mathcal{D}_S$, Similarity matrix S, Selection sizes k_1, k_2
Ensure: Selected target set $\mathcal{D}_T$ and source set $\mathcal{S}$

Stage 1: Facility Location
$\mathcal{T} \leftarrow \emptyset$
while $|\mathcal{T}| < k_1$ **do**
  $i^* \leftarrow \arg\max_{i \in \mathcal{D}_{\mathcal{T}} \setminus \mathcal{T}} f(\mathcal{T} \cup \{i\}) - f(\mathcal{T})$
  $\mathcal{T} \leftarrow \mathcal{T} \cup \{i^*\}$
end while
Stage 2: Mutual Information
$\mathcal{S} \leftarrow \emptyset$
while $|\mathcal{S}| < k_2$ **do**
  $j^* \leftarrow \arg\max_{j \in \mathcal{D}_S \setminus \mathcal{S}} f(\mathcal{S} \cup \{j\}; \mathcal{T}) - f(\mathcal{S}; \mathcal{T})$
  $\mathcal{S} \leftarrow \mathcal{S} \cup \{j^*\}$
end while
return $\mathcal{T} \oplus \mathcal{S}$

3 Experimental Setup

In this section we give a detailed description of the experimental settings. We follow some of the state of the art research done in domain adaptation.

Datasets

We use Amazon reviews data as our source domain and yelp reviews as our target domain. We believe that this dataset is very divergent in terms of data. We reduce them to a binary classification by setting any review 3 and below as negative and anything above 3 as positive. This is a method that was used by (Ben-David et al., 2022).

Model Architecture and Training

We use RoBERTa-base (125M parameters) (Liu et al., 2019) as our foundation model for all experiments. We use the same settings in (Ben-David et al., 2022). All our experiments are conducted on a single Nvidia A5500 24GB GPU, 128GB RAM computer. We set the model parameters as follow, we use a learning rate of $2e-5$ with cosine decay, we set the number of epochs to 3, and we use all-MiniLM-L6-v2 to embed our prompts.

3.1 Evaluation Metrics

We evaluate our approach using metrics that capture task performance, domain adaptation capability, and computational efficiency. Task performance is assessed using the F1 score, which balances precision and recall, as well as overall prediction accuracy on the fine-tuned task. To measure efficiency, we analyze the model's convergence rate, reflecting how quickly it learns from the training data. Together, these metrics provide a comprehensive assessment of the proposed method's effectiveness in addressing the target problem.

4 Experimental Results

In this section we present the results from our experiments.

4.1 RQ1: Target Domain Sample Selection

To address **RQ1**, we conduct a controlled experiment using 500 data points for fine-tuning a sentiment analysis model. We evaluate three different data selection strategies for incorporating target domain samples:

No Sampling. In this baseline setting, no samples from the target domain are included. We randomly select 500 data points from the source domain and use them to fine-tune the model.

Random Sampling. We randomly select 50 data points from the target domain and combine them with 450 randomly selected data points from the

source domain. The combined dataset of 500 samples is used for fine-tuning.

FL Sampling. We use the Facility Location function to select 50 representative data points from the target domain. These are combined with 450 randomly sampled points from the source domain to form the fine-tuning dataset.

Analysis of Convergence. Figure 2 shows the training loss curves for our three experimental setups addressing RQ1. From the figure, we see that No sampling converges faster. This is because the data is all from the same source making it easier for the model to learn.

FL Sampling, which leverages a facility location function to select representative samples, consistently achieves lower training loss than Random Sampling. Its sharper initial decline suggests more efficient learning, due to the higher informativeness of the selected samples and how closely they are related to the majority source domain selected for the downstream task. By the final epoch, the persistent gap in loss confirms that submodular selection leads to more effective adaptation than random sampling, despite using the same number of training examples.

4.2 RQ2: Feature Importance in Sample Selection

In this section, we want to study the nature of the selected data points in relation to the entire dataset. To understand how features in the annotated samples guide source domain selection, we analyzed the embedding space using t-SNE visualization. Figure 4 compares the distribution of data points selected by DASS versus random selection.

The left plot (Figure 4a) shows that randomly selected source domain samples (red points) are distributed throughout the embedding space without clear alignment to the target domain (green points). In contrast, the right plot (Figure 4b) demonstrates that samples selected by our DASS framework cluster more closely with target domain data.

This visualization supports our hypothesis that DASS effectively identifies source domain examples whose feature representations are more similar to the target domain. The mutual information function successfully captures semantic relationships between domains, enabling more efficient transfer learning by focusing on samples most relevant to the target task.

Table 1: Classification accuracy on the Yelp review target domain test set

Method	Accuracy	AUC
Source Only	0.876	0.955
FL-Random	0.876	0.927
Random-DASS	0.887	0.952
DASS (Our Approach)	0.883	0.958

4.3 RQ3: Effectiveness of the Complete Framework

In this section, we test the performance of a model trained on the data selected using our framework against other benchmarks. We perform three basic experiments using RoBERTa-base (125M parameters) (Liu et al., 2019) and finetune it with the data. We set the budget to 1000, with 30% of the data points coming from the target domain. We check the performance of the model using the AUC score following the settings of (Ben-David et al., 2022).

Source Only. In this baseline, we randomly sample a subset of the source domain data—equal in size to the annotation budget—and use it to fine-tune the model. This setting serves to highlight the necessity of domain adaptation by demonstrating the performance degradation that occurs when a model trained exclusively on data from one domain is directly applied to a different target domain.

FL-Random. This experiment combines a carefully selected subset from the target domain—obtained using the facility location function—with a randomly sampled subset from the source domain. The combined dataset is used to fine-tune the model. The purpose of this setting is to assess the impact of naively augmenting high-quality target domain data with unrelated source domain data, without regard to the data distribution. It serves as a benchmark to emphasize the importance of our framework's second stage, which incorporates Submodular Mutual Information (SMI) for source data selection.

Random-DASS. In this variant, we randomly sample target domain data for annotation. These selected points are then used in the first stage of our framework to guide the selection of source domain data using Submodular Mutual Information. This setup is designed to isolate and evaluate the contribution of the first stage of our framework, helping to determine whether SMI-based source selection

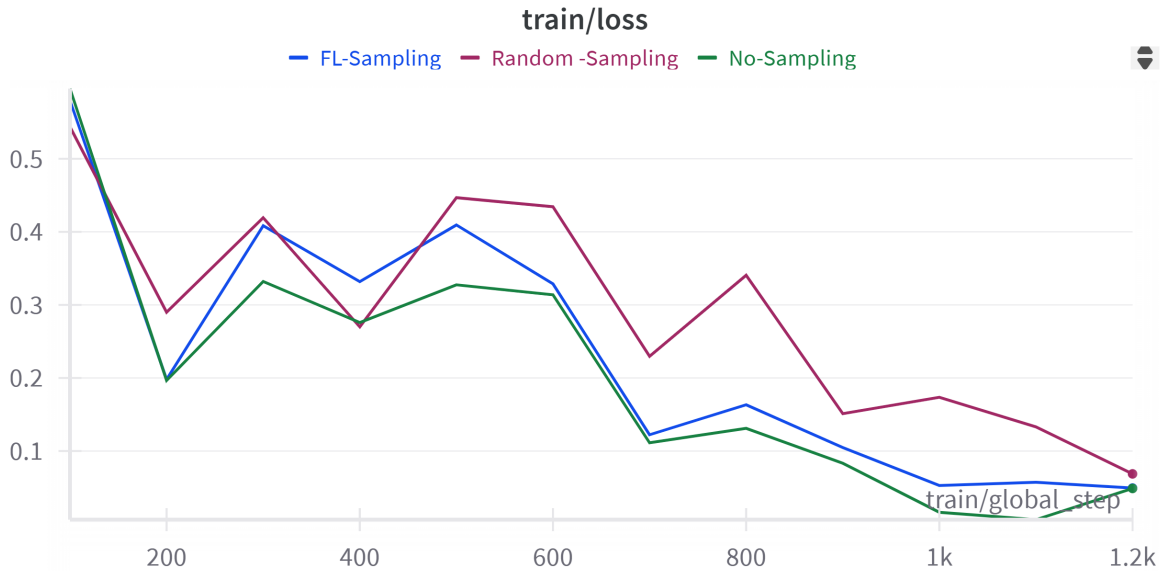


Figure 2: Model Convergence During Training

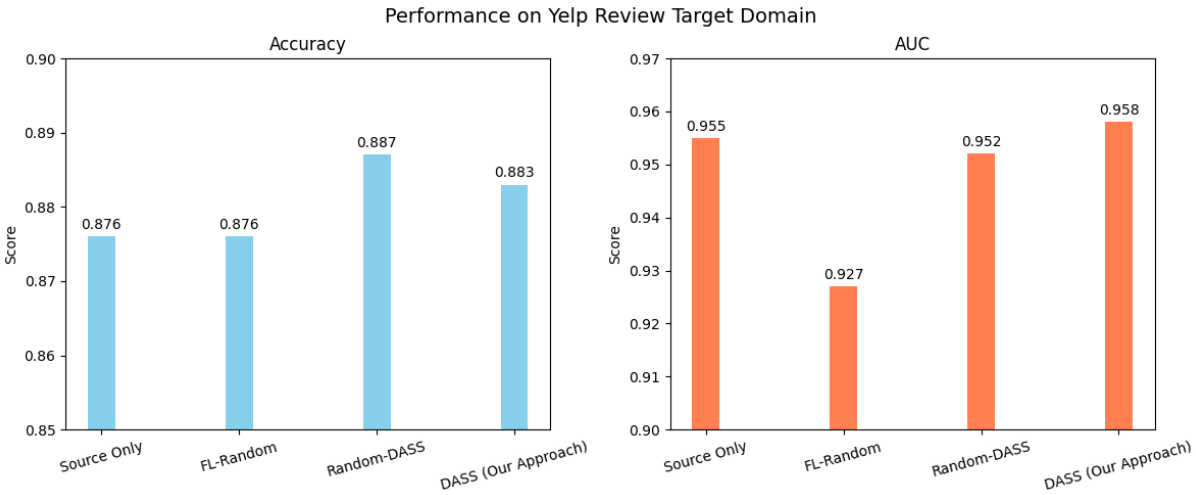
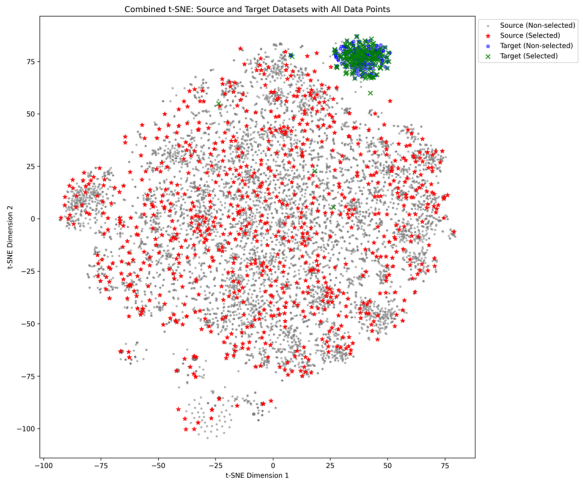
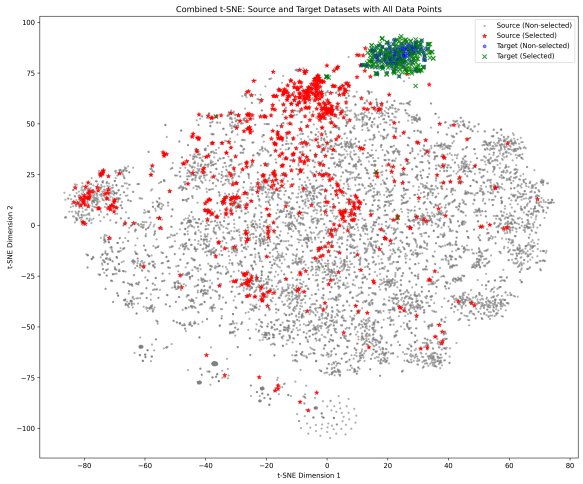


Figure 3: Accuracy-AUC Scores



(a) TSNE-plot of data selected using Random Selection



(b) TSNE-plot of data selected using our Framework

Figure 4: Comparison between DASS and Random Selection

provides benefits even when target domain data is randomly chosen.

DASS. Our proposed framework, **DASS** (Domain-Aware Submodular Selection), begins by selecting a representative subset from the target domain using the facility location function. These selected points are then used to guide source domain data selection via the Facility Location Mutual Information (FLMI) function. The resulting fine-tuning dataset consists of both representative target domain samples and a complementary, distribution-aware subset of source domain data. This two-stage approach aims to maximize cross-domain alignment and downstream task performance.

Results. Table 1 summarizes the performance of all methods on the Yelp review target domain test set. Our proposed framework, DASS, achieves the highest AUC score of 0.958, outperforming all baseline methods. It also attains a strong accuracy of 0.883, confirming the benefit of jointly optimizing target and source data selection via submodular objectives.

The Source Only baseline, which fine-tunes the model using a randomly sampled subset from the source domain, yields an AUC of 0.955 and an accuracy of 0.876, illustrating the limitations of relying solely on out-of-domain data.

In the FL-Random setting, where target domain samples are selected using facility location and source samples are randomly selected, performance drops slightly to an AUC of 0.927, despite having high-quality target points. This emphasizes the importance of selecting source data that is distributionally aligned with the target.

The Random-DASS variant—where target data is selected randomly but source data is chosen using our submodular mutual information approach—achieves an AUC of 0.952 and accuracy of 0.887, highlighting the standalone benefit of intelligent source data selection. However, its performance remains slightly lower than DASS, indicating that high-quality target data is also critical for guiding source selection effectively.

Overall, these results demonstrate that the full DASS framework—by integrating structured target data selection with submodular source selection—achieves the best trade-off between accuracy and robustness. The results validate the contribution of both stages in our approach and highlight its effectiveness in low-resource domain adaptation settings.

5 Related Works

This section covers relevant literature in data subset selection, domain adaptation techniques, and the application of submodular optimization in machine learning tasks.

5.1 Domain Adaptation Approaches

Domain adaptation addresses the challenge of transferring knowledge from source domains to target domains with different distributions. Kim et al. (Kim et al., 2024) proposed a meta-reweighting mechanism that dynamically selects data points to reduce loss values during training, though this approach incurs significant computational costs.

In visual domains, Rangwani et al. (Rangwani et al., 2021) introduced a submodular adversarial approach for image processing tasks, but its application to natural language tasks remains largely unexplored. More relevant to our work, Shinohara et al. (Shinohara and Watanabe, 2023) applied submodular optimization to Text-to-Speech systems by minimizing KL-divergence between selected data points and the target domain, though they did not investigate scenarios where labeled target domain data could guide the training process—a key distinction in our approach.

5.2 Data Subset Selection Techniques

The quality of training data significantly impacts model performance, making effective data subset selection crucial in machine learning applications (Kim et al., 2024; Ghorbani and Zou, 2019). These techniques have been successfully deployed across diverse domains including speech recognition (Kim et al., 2024), natural language processing (Renduchintala et al., 2023; Mavromatis et al., 2023; Renduchintala et al., 2024), and computer vision (Birodkar et al., 2019), with recent work by Renduchintala et al. (Renduchintala et al., 2024) demonstrating that intelligently selected subsets can sometimes outperform training on complete datasets, highlighting the importance of quality over quantity in data selection.

5.3 Submodular Optimization in Machine Learning

Submodular functions exhibit a diminishing returns property that makes them particularly suitable for optimization problems involving set selection (Bach, 2011), providing efficient algorithms for identifying near-optimal subsets even from extremely large candidate sets. In machine learning contexts, submodular optimization has been applied to various tasks including active learning (Wei et al., 2014; Renduchintala et al., 2023), where it helps identify the most informative samples for annotation, with promising applications in speech recognition (Kim et al., 2024) and image classification (Rangwani et al., 2021), though its use in text-based domain adaptation remains relatively underexplored. Our work extends these efforts by formulating data similarity as a submodular optimization problem, enabling us to select subsets that effectively represent the target distribution while incorporating insights from partially labeled target domain data—bridging the gap between unsupervised domain adaptation and supervised learning.

6 Conclusion & Future Work

This work provides a framework for domain adaptation in NLP using submodular information theoretical approaches with proven algorithmic guarantees (Iyer et al., 2021). Our experimental results demonstrate that submodular functions effectively capture the relationship between source and target domains while maintaining computational efficiency. The proposed framework consistently outperforms existing methods across multiple benchmark datasets, particularly in low-resource scenarios where traditional approaches often struggle. By formalizing domain adaptation through the lens of submodular optimization, we establish a theoretically sound foundation that unifies several previously disparate approaches to transfer learning in natural language processing.

6.1 Limitations

While our results are promising, there are several limitations to this study. First, the experiments were conducted on a relatively small dataset, which may limit the generalizability of our findings. Future work should validate the framework on larger and more diverse datasets to ensure broader applicability. Second, our evaluation was restricted to binary sentiment classification, a relatively simple task. Extending the approach to more complex scenarios, such as multi-class sentiment classification or other NLP tasks, would provide a more rigorous test of its effectiveness. Finally, while we demonstrate quantitative improvements, deeper qualitative and error analyses could offer additional insights into how and why the framework performs as it does, and where it might still fall short.

6.2 Future Directions

Our work opens several promising avenues for future research.

Multi-domain adaptation. One potential direction is extending DASS to settings involving multiple source and target domains. In such scenarios, hierarchical submodular functions could be employed to better model inter-domain relationships, enabling more effective and structured data selection across diverse domains.

Alternative submodular functions. Another area for exploration involves the use of alternative submodular functions tailored to specific NLP tasks. For instance, log-determinant functions could be particularly advantageous for promoting semantic diversity, which is often critical for tasks requiring broad linguistic coverage.

Integration with representation learning. There is also significant promise in integrating DASS with domain-invariant representation learning techniques. Such a combination could offer complementary benefits, enhancing both data selection and feature alignment to improve model generalization across domains.

Complex NLP tasks. Applying our framework to more complex NLP tasks—such as machine translation, summarization, and question answering—is another compelling direction. These tasks introduce distinct structural and linguistic challenges, offering opportunities to design specialized submodular objectives that are better suited to their requirements.

Dynamic Adaptation. Developing dynamic or online variants of DASS presents an important research opportunity. These methods would enable continuous adaptation as new target domain data becomes available, making the framework more applicable in real-world settings where data evolves over time.

By establishing a theoretical connection between submodular optimization and domain adaptation, we provide a foundation for more efficient and effective transfer learning in natural language processing. This is particularly valuable in low-resource scenarios where labeled data is scarce and annotation costs are high.

References

Hema Ala, Vandan Mujadia, and Dipti Sharma. 2021. Domain adaptation for Hindi-Telugu machine translation using domain specific back translation. In *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2021)*, pages 26–34, Held Online. INCOMA Ltd.

Francis R. Bach. 2011. Learning with submodular functions: A convex optimization perspective. *CoRR*, abs/1111.6453.

Eyal Ben-David, Yftah Ziser, and Roi Reichart. 2022. Domain adaptation from scratch.

Shai Ben-David, John Blitzer, Koby Crammer, Alex Kulesza, Fernando Pereira, and Jennifer Wortman Vaughan. 2010. A theory of learning from different domains. *Machine Learning*, 79(1-2):151–175.

Vighnesh Birodkar, Hossein Mobahi, and Samy Bengio. 2019. Semantic redundancies in image-classification datasets: The 10% you don't need. *CoRR*, abs/1901.11409.

Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. 2015. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 632–642, Lisbon, Portugal. Association for Computational Linguistics.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are few-shot learners.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, Dasha Valter, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Zhao, Yanping Huang, Andrew Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. 2022. Scaling instruction-finetuned language models.

Alexandre Diniz da Costa, Mateus Coutinho Marim, Ely Matos, and Tiago Timponi Torrent. 2022. Domain adaptation in neural machine translation using a qualia-enriched FrameNet. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 1–12, Marseille, France. European Language Resources Association.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Satoru Fujishige. 2005. *Submodular functions and optimization*, 2nd ed edition. Number 58 in Annals of discrete mathematics. Elsevier, Amsterdam Boston Oxford.

Amirata Ghorbani and James Zou. 2019. Data shapley: Equitable valuation of data for machine learning. In *International conference on machine learning*, pages 2242–2251. PMLR.

Xu Guo and Han Yu. 2022. On the Domain Adaptation and Generalization of Pretrained Language Models: A Survey. ArXiv:2211.03154 [cs].

Rishabh Iyer, Ninad Khargonkar, Jeff Bilmes, and Himanshu Asnani. 2021. Submodular combinatorial information measures with applications in machine learning.

Zixuan Ke, Mohammad Kachuee, and Sungjin Lee. 2022. Domain-aware contrastive knowledge transfer for multi-domain imbalanced data. In *Proceedings of the 12th Workshop on Computational Approaches to Subjectivity, Sentiment & Social Media Analysis*, pages 25–36, Dublin, Ireland. Association for Computational Linguistics.

Junghun Kim, Ka Hyun Park, Hoyoung Yoon, and U Kang. 2024. Domain-Aware Data Selection for Speech Classification via Meta-Reweighting. In *Interspeech 2024*, pages 797–801. ISCA.

Suraj Kothawade, Vishal Kaushal, Ganesh Ramakrishnan, Jeff Bilmes, and Rishabh Iyer. 2022. Prism: A rich class of parameterized submodular information measures for guided subset selection.

Andreas Krause and Daniel Golovin. 2014. Submodular Function Maximization. In Lucas Bordeaux, Youssef Hamadi, and Pushmeet Kohli, editors, *Tractability*, 1 edition, pages 71–104. Cambridge University Press.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized BERT pretraining approach. *CoRR*, abs/1907.11692.

Mitchell P. Marcus, Beatrice Santorini, and Mary Ann Marcinkiewicz. 1993. Building a large annotated corpus of English: The Penn Treebank. *Computational Linguistics*, 19(2):313–330.

Costas Mavromatis, Balasubramaniam Srinivasan, Zhengyuan Shen, Jiani Zhang, Huzefa Rangwala, Christos Faloutsos, and George Karypis. 2023. Which Examples to Annotate for In-Context Learning? Towards Effective and Efficient Selection. ArXiv:2310.20046 [cs].

Baharan Mirzasoleiman, Ashwinkumar Badanidiyuru, Amin Karbasi, Jan Vondrák, and Andreas Krause. 2015. Lazier than lazy greedy. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 29.

Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. 2018. Improving language understanding by generative pre-training.

Harsh Rangwani, Arihant Jain, Sumukh K Aithal, and R. Venkatesh Babu. 2021. S^3 VAADA: Submodular Subset Selection for Virtual Adversarial Active Domain Adaptation. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7496–7505, Montreal, QC, Canada. IEEE.

H. S. V. N. S. Kowndinya Renduchintala, Sumit Bhatia, and Ganesh Ramakrishnan. 2024. SMART: Submodular Data Mixture Strategy for Instruction Tuning. ArXiv:2403.08370 [cs].

H. S. V. N. S. Kowndinya Renduchintala, Krishnateja Killamsetty, Sumit Bhatia, Milan Aggarwal, Ganesh Ramakrishnan, Rishabh Iyer, and Balaji Krishnamurthy. 2023. INGENIOUS: Using Informative Data Subsets for Efficient Pre-Training of Language Models. ArXiv:2305.06677 [cs].

Danielle Saunders. 2022. Domain adaptation for neural machine translation. In *Proceedings of the 23rd Annual Conference of the European Association for Machine Translation*, pages 9–10, Ghent, Belgium. European Association for Machine Translation.

Burr Settles. 2009. Active Learning Literature Survey.

Yusuke Shinohara and Shinji Watanabe. 2023. Domain Adaptation by Data Distribution Matching Via Submodularity For Speech Recognition. In *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1–7, Taipei, Taiwan. IEEE.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. Llama: Open and efficient foundation language models.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.

Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. 2022. Finetuned language models are zero-shot learners.

Kai Wei, Yuzong Liu, Katrin Kirchhoff, Chris Bartels, and Jeff Bilmes. 2014. Submodular subset selection for large-scale speech training data. In *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 3311–3315, Florence, Italy. IEEE.

Garrett Wilson and Diane J. Cook. 2020. A Survey of Unsupervised Deep Domain Adaptation. *ACM Transactions on Intelligent Systems and Technology*, 11(5):1–46.